EPSR-test-ex121 Bragg peaks with Lorentzian broadening, hklqmin cut, triclinic crystal

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the Bragg-peak calculation of the `epsr` engine [1,2,3] for a general triclinic lattice, with Lorentzian peak shapes and a lower Q limit for broadening.

Code exercised

Fortran (original, fortran_src/):

- `ato_manipulation_rev4.f`: `bragginten` (reciprocal lattice, hkl list, `.bragg`), `crystfrac`
- `epsr_standalone_rev3.f`: `runepsr17` (keywords; `.inten`, Debye-Waller factor, truncation, replacement of the partial S(Q))
- `ato_manipulation_rev4.f`: `atogen`; `epsr_standalone_rev3.f`: `gensq`

Rust (port, rust/epsr-core/src/):

- `bragg.rs`: `Bragg::read_keywords`, `bragginten`, `structure_and_lattice`, `bragg_peaks`
- `structure.rs`: `atogen`, `gensq`; `epsr.rs`: `Epsr::iterate`

Physical system

A triclinic crystal of tethered one-atom molecules built with `makelattice` from a cell a, b, c = 4.3, 4.8, 5.4 Å, alpha, beta, gamma = 80°, 100°, 110° with S at (0,0,0) and O1 at (0.5,0.45,0.55) (no charges), replicated 7 × 7 × 6: 588 atoms in a 30.1 × 33.6 × 32.4 Å box at 300 K. The labels S and O1 exist in the EPSR-test `.pcof` and weights, which are used unchanged (only the S-S, S-O1, O1-O1 pairs are matched); the data are numerical input only. For a crystal the simulation box holds only a few unit cells, so the Bragg part of the structure factor [4] is computed from the reciprocal lattice: intensities per atom pair and reflection are broadened into peaks, divided by the number of unit cells, damped by a Debye-Waller factor and replace the intermolecular partial S(Q) up to `qbraggmax`. A negative width selects Lorentzian peaks.

Inputs

`scripts/make_case.sh` (with `make_triclinic_crystal.sh`, `setkw.py`) regenerates the inputs, verified byte-identical. Keywords changed: `iinit 1`, `ntimes 2`, `qwidthqmax -0.02 4.0`, `mplicities 7 7 6 1 1 2` (6 words; the value 2 is reset to 1), `hklqmin 1.0 8.0 0.0` (no broadening below 1 1/Å, `rminbragg` 8 Å, no Debye-Waller factor). `commands.txt`: `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Exercised: triclinic reciprocal lattice with non-zero off-diagonal components and k/l ranges depending on h and k, the Lorentzian branch, the `hklqmin` cut, the `debyewaller = 0` branch, tethered molecules in `crystfrac` and a non-cubic box. `.bragg`, `.inten` and all other outputs are identical in both passes.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] G. L. Squires, Introduction to the Theory of Thermal Neutron Scattering, Cambridge University Press (1978).
[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.